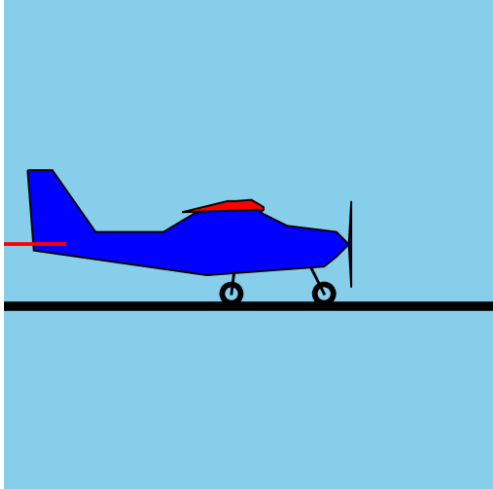


1 Test 0

```
test_aircraft0 = TecnamP92()  
simulate_one_time_step(test_aircraft0 , 0.01)
```



Engine torque generated

```
 $\tau_e = \text{TecnamP92.engine\_torque}(\textit{throttle}, N_e)$   
= TecnamP92.engine_torque(0, 1000)  
= 5.87018738
```

Airspeed

```
 $|v| = |(0, 0)|$   
= 0.0
```

Engine RPM

$$N_e = 1000$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{1000}{2.43} \\ &= 411.52263374485597 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{0.0}{\frac{411.52263374485597}{60} \cdot 1.74} \\ &= 0.0 \end{aligned}$$

Air Density

$$h = 0$$

$$\begin{aligned} T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 0 \\ &= 288.15 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{288.15}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 101325.0 \\ \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{101325.0}{287.05 \cdot 288.15} \\ &= 1.2250122659906946 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{411.52263374485597 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 43.094549431958754 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{TecnamP92.propeller_torque_coefficient}(0.0) \cdot 1.2250122659906946 \cdot +43.094549431958754^2 \cdot 1.74^5 \\
&= 0.0002990229885057471 \cdot 1.2250122659906946 \cdot +43.094549431958754^2 \cdot 1.74^5 \\
&= 10.850155068063646
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{10.850155068063646}{2.43} \\
&= 4.465084390149649
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= 5.87018738 - 4.465084390149649 \\
&= 1.4051029898503513
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{1.4051029898503513}{0.1} \\
&= 14.051029898503513
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 1000 + 14.051029898503513 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 1002.6835490366547
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 0 + 1.5707963267948966 \\
&= 1.5707963267948966
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 0 + 0.03490658503988659 \\
&= 0.03490658503988659
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (0, 0) \\
&= 0.0
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.03490658503988659 - 0.0 \\
&= 0.03490658503988659
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.03490658503988659) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 12.2 \cdot 0.56 \\
&= 0.0
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0 \text{ in direction normal to } 0.0 \\
&= 0.0 \text{ in direction } 1.5707963267948966 \\
&= (0.0, 0.0)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$|F_i| = 0 \text{ since aircraft is stationary}$$

Induced Drag (wing)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0 \text{ in direction away from } 0.0 \\ &= (0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 1.5707963267948966 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 3.141592653589793 \\ &= -0.0 \end{aligned}$$

Compute Moment(wing)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -0.0 \cdot 0.54 \\ &= -0.0 \end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
 \gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
 &= 0 + 3.141592653589793 \\
 &= 3.141592653589793
 \end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
 \text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
 &= 0 + 0 \\
 &= 0
 \end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
 \phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
 &= 0 - 0.0 \\
 &= 0.0
 \end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
 |F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
 &= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(0.0) \\
 &= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 1.97 \cdot 0.0 \\
 &= 0.0
 \end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
 F_L &= |F_L| \text{ in direction normal to relative airflow} \\
 &= 0.0 \text{ in direction normal to } 0.0 \\
 &= 0.0 \text{ in direction } 1.5707963267948966 \\
 &= (0.0, 0.0)
 \end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$|F_i| = 0 \text{ since aircraft is stationary}$$

Induced Drag (stabilizer)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0 \text{ in direction away from } 0.0 \\ &= (0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 3.141592653589793 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 4.71238898038469 \\ &= -0.0 \end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -0.0 \cdot 4 \\ &= -0.0 \end{aligned}$$

Net Moment

$$\begin{aligned} M_{net} &= M_w + M_s \\ &= -0.0 + -0.0 \\ &= -0.0 \end{aligned}$$

Angular Acceleration

$$\begin{aligned} \alpha_t &= \frac{M_{net}}{I_p} \\ &= \frac{-0.0}{1000.0} \\ &= -0.0 \end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned} \omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\ &= 0 + -0.0 \cdot 0.01 \\ &= 0.0 \end{aligned}$$

$$\begin{aligned} \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\ &= 0 + 0.0 \cdot 0.01 \\ &= 0.0 \end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.0) \cdot 1.2250122659906946 \cdot 43.21019576857195^2 \cdot 1.74^4 \\
&= 0.00435 \cdot 1.2250122659906946 \cdot 43.21019576857195^2 \cdot 1.74^4 \\
&= 91.20090653921454 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 91.20090653921454 \text{ Newtons, in direction } 0.0 \\
&= (91.20090653921454, 0.0)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 0.03 \cdot 1.2 \\
&= 0.0 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 0.0 \text{ Newtons in direction away from } 0.0 \\
&= (-0.0, -0.0)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$\begin{aligned}
|F_r| &\leq \mu_r \cdot |F_g| \\
&= 0.04 \cdot -5880.0 \\
&= -235.20000000000002 \\
F_r &= (0, 0) \text{ since not on ground or not moving}
\end{aligned}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (91.20090653921454, 0.0) + (0.0, 0.0) + \\
&\quad (0.0, 0.0) + (-0.0, -0.0) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (91.20090653921454, -5880.0)
\end{aligned}$$

Linear Acceleration

$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(91.20090653921454, -5880.0)}{600.0} \\&= (0.1520015108986909, -9.8)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (0, 0) + (0.1520015108986909, -9.8) \cdot 0.01 \\&= (0.0015200151089869091, -0.098) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (0, 0) + (0.0015200151089869091, -0.098) \cdot 0.01 \\&= (1.5200151089869091e-05, -0.00098)\end{aligned}$$

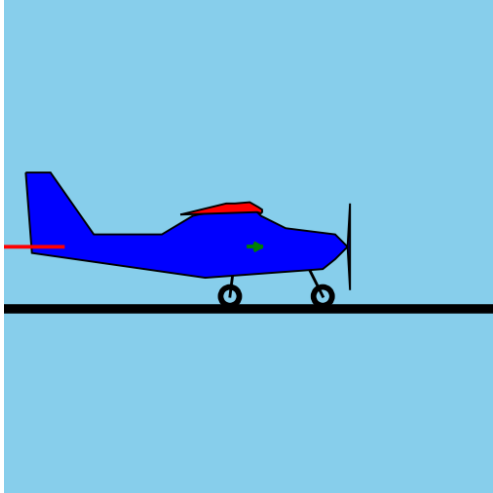
Correct for on ground

position corrected from $(1.5200151089869091e-05, -0.00098)$ to $(1.5200151089869091e-05, 0)$

linear velocity corrected from $(0.0015200151089869091, -0.098)$ to $(0.0015200151089869091, 0)$

2 Test 1

```
test_aircraft1 = TecnamP92(angular_velocity=9.462226495761285e-05, engine_rpm=5324.945824260662,  
    linear_velocity=(2.5047676762937745, 0), pitch_angle_radians=1.6090582357960263e-05, position=(0.8547359555756816, 0),  
    throttle=1.0, time_elapsed=1.8)  
simulate_one_time_step(test_aircraft1, 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine_torque}(\text{throttle}, N_e) \\ &= \text{TecnamP92.engine_torque}(1.0, 5324.945824260662) \\ &= 126.71209985095138\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(2.5047676762937745, 0)| \\ &= 2.5047676762937745\end{aligned}$$

Engine RPM

$$N_e = 5324.945824260662$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{5324.945824260662}{2.43} \\ &= 2191.3357301484207 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{2.5047676762937745}{\frac{2191.3357301484207}{60} \cdot 1.74} \\ &= 0.039414909360645504 \end{aligned}$$

Air Density

$$h = 0$$

$$\begin{aligned} T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 0 \\ &= 288.15 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{288.15}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 101325.0 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{101325.0}{287.05 \cdot 288.15} \\ &= 1.2250122659906946 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{2191.3357301484207 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 229.47614104610344 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{TecnamP92.propeller_torque_coefficient}(0.039414909360645504) \cdot 1.2250122659906946 \cdot +229.47614104610344^2 \cdot 1.74^5 \\
&= 0.00029893370488046506 \cdot 1.2250122659906946 \cdot +229.47614104610344^2 \cdot 1.74^5 \\
&= 307.56480659405247
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{307.56480659405247}{2.43} \\
&= 126.5698792568117
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= 126.71209985095138 - 126.5698792568117 \\
&= 0.14222059413967258
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{0.14222059413967258}{0.1} \\
&= 1.4222059413967258
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 5324.945824260662 + 1.4222059413967258 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 5325.217445587464
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 1.6090582357960263e - 05 + 1.5707963267948966 \\
&= 1.5708124173772544
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 1.6090582357960263e - 05 + 0.03490658503988659 \\
&= 0.03492267562224455
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (2.5047676762937745, 0) \\
&= 0.0
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.03492267562224455 - 0.0 \\
&= 0.03492267562224455
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.03492267562224455) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 12.2 \cdot 0.5600737537967215 \\
&= 26.257319806705347
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 26.257319806705347 \text{ in direction normal to } 0.0 \\
&= 26.257319806705347 \text{ in direction } 1.5707963267948966 \\
&= (1.6077971327735051e - 15, 26.257319806705347)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned}|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\&= \frac{26.257319806705347^2}{0.5 \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\&= 0.8253719185357837\end{aligned}$$

Induced Drag (wing)

$$\begin{aligned}F_i &= |F_i| \text{ in direction away from relative airflow} \\&= 0.8253719185357837 \text{ in direction away from } 0.0 \\&= (-0.8253719185357837, -0.0)\end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned}F_a &= F_L + F_i \\&= (1.6077971327735051e-15, 26.257319806705347) + (-0.8253719185357837, -0.0) \\&= (-0.8253719185357822, 26.257319806705347)\end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned}|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\&= \text{component of } (-0.8253719185357822, 26.257319806705347) \text{ in direction } 1.5708124173772544 + 90^\circ \\&= \text{component of } (-0.8253719185357822, 26.257319806705347) \text{ in direction } 3.141608744172151 \\&= 0.8249494228621047\end{aligned}$$

Compute Moment(wing)

$$\begin{aligned}M &= |F_n| \cdot d \\&= 0.8249494228621047 \cdot 0.54 \\&= 0.44547268834553655\end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 1.6090582357960263e - 05 + 3.141592653589793 \\
&= 3.141608744172151
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 1.6090582357960263e - 05 + 0 \\
&= 1.6090582357960263e - 05
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= 1.6090582357960263e - 05 - 0.0 \\
&= 1.6090582357960263e - 05
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(1.6090582357960263e - 05) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 1.97 \cdot 9.219224590187834e - 05 \\
&= 0.0006979205128584067
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0006979205128584067 \text{ in direction normal to } 0.0 \\
&= 0.0006979205128584067 \text{ in direction } 1.5707963267948966 \\
&= (4.273530610656634e - 20, 0.0006979205128584067)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{0.0006979205128584067^2}{0.5 \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\
&= 7.9959388715436e - 09
\end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 7.9959388715436e - 09 \text{ in direction away from } 0.0 \\
&= (-7.9959388715436e - 09, -0.0)
\end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (4.273530610656634e - 20, 0.0006979205128584067) + (-7.9959388715436e - 09, -0.0) \\
&= (-7.995938871500865e - 09, 0.0006979205128584067)
\end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-7.995938871500865e - 09, 0.0006979205128584067) \text{ in direction } 3.141608744172151 + 90^\circ \\
&= \text{component of } (-7.995938871500865e - 09, 0.0006979205128584067) \text{ in direction } 4.712405070967048 \\
&= -0.0006979205128967178
\end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= -0.0006979205128967178 \cdot 4 \\
&= -0.0027916820515868713
\end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= 0.44547268834553655 + -0.0027916820515868713 \\&= 0.4426810062939497\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{0.4426810062939497}{1000.0} \\&= 0.0004426810062939497\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= 9.462226495761285e-05 + 0.0004426810062939497 \cdot 0.01 \\&= 9.904907502055234e-05 \\\theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= 1.6090582357960263e-05 + 9.904907502055234e-05 \cdot 0.01 \\&= 1.7081073108165786e-05\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\&= \text{TecnamP92.propeller_thrust_coefficient}(0.03941289893945883) \cdot 1.2250122659906946 \cdot 229.48784644479804^2 \cdot 1.74^4 \\&= 0.004343242811777768 \cdot 1.2250122659906946 \cdot 229.48784644479804^2 \cdot 1.74^4 \\&= 2568.448847895754 \\F_t &= |F_t| \text{ in direction } \theta \\&= 2568.448847895754 \text{ Newtons, in direction } 1.7081073108165786e - 05 \\&= (2568.448847521065, 0.043871862543358196)\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\&= 0.5 \cdot 1.2250122659906946 \cdot 2.5047676762937745^2 \cdot 0.03 \cdot 1.2 \\&= 0.13834002271634122 \\F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\&= 0.13834002271634122 \text{ Newtons in direction away from } 0.0 \\&= (-0.13834002271634122, -0.0)\end{aligned}$$

Weight Force

$$\begin{aligned}F_g &= (0, -m \cdot g) \\&= (0, -600.0 \cdot 9.8) \\&= (0, -5880.0)\end{aligned}$$

Rolling Resistance Force

$$\begin{aligned}|F_r| &\leq \mu_r \cdot |F_g| \\&= 0.04 \cdot -5880.0 \\&= -235.20000000000002 \\F_r &= |F_r| \text{ in direction opposite horizontal direction of travel} \\&= (-235.20000000000002, 0)\end{aligned}$$

Net Force

$$\begin{aligned}F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\&= (2568.448847521065, 0.043871862543358196) + (-0.8253852998653534, 26.257532654039995) + \\&\quad (-9.010652539614161e - 09, 0.0007408825261023188) + (-0.13834002271634122, -0.0) + \\&\quad (0, -5880.0) + (-235.20000000000002, 0) \\&= (2332.2851221894725, -5853.69785460089)\end{aligned}$$

Linear Acceleration

$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(2332.2851221894725, -5853.69785460089)}{600.0} \\&= (3.8871418703157876, -9.756163091001484)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (2.5047676762937745, 0) + (3.8871418703157876, -9.756163091001484) \cdot 0.01 \\&= (2.5436390949969323, -0.09756163091001485) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (0.8547359555756816, 0) + (2.5436390949969323, -0.09756163091001485) \cdot 0.01 \\&= (0.8801723465256509, -0.0009756163091001486)\end{aligned}$$

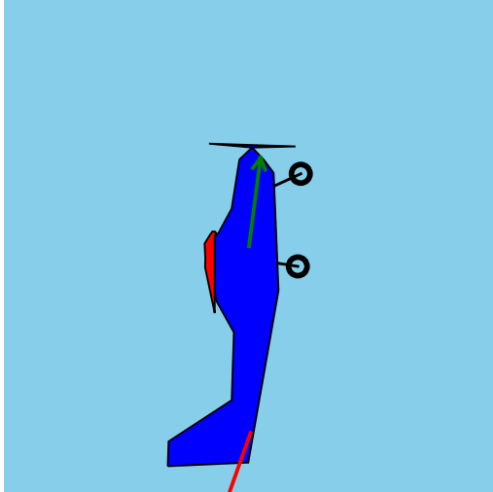
Correct for on ground

position corrected from $(0.8801723465256509, -0.0009756163091001486)$ to $(0.8801723465256509, 0)$

linear velocity corrected from $(2.5436390949969323, -0.09756163091001485)$ to $(2.5436390949969323, 0)$

3 Test 2

```
test_aircraft2 = TecnamP92(angular_velocity=0.1645214403696835, engine_rpm=5491.802723731343,  
    linear_velocity=(2.3292949232141194, 17.899852673407583), pitch_angle_radians=1.5387410399488646,  
    position=(1797.8474288628177, 514.0760387844257), stabilizer_angle_radians=-0.10297442586766545, throttle=1.0,  
    time_elapsed=58.197)  
simulate_one_time_step(test_aircraft2 , 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine_torque}(\text{throttle}, N_e) \\ &= \text{TecnamP92.engine_torque}(1.0, 5491.802723731343) \\ &= 126.17328228000642\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(2.3292949232141194, 17.899852673407583)| \\ &= 18.050771190423074\end{aligned}$$

Engine RPM

$$N_e = 5491.802723731343$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{5491.802723731343}{2.43} \\ &= 2260.0011208770957 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{18.050771190423074}{\frac{2260.0011208770957}{60} \cdot 1.74} \\ &= 0.27541596334920915 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 514.0760387844257 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 514.0760387844257 \\ &= 284.8085057479012 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{284.8085057479012}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 95303.72052918143 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{95303.72052918143}{287.05 \cdot 284.8085057479012} \\ &= 1.1657336923917203 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{2260.0011208770957 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 236.66676394840607 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{TecnamP92.propeller_torque_coefficient}(0.27541596334920915) \cdot 1.1657336923917203 \cdot +236.66676394840607^2 \cdot 1.74^5 \\
&= 0.00029466356592715103 \cdot 1.1657336923917203 \cdot +236.66676394840607^2 \cdot 1.74^5 \\
&= 306.8644477206299
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{306.8644477206299}{2.43} \\
&= 126.28166572865429
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= 126.17328228000642 - 126.28166572865429 \\
&= -0.10838344864787075
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{-0.10838344864787075}{0.1} \\
&= -1.0838344864787075
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 5491.802723731343 + -1.0838344864787075 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 5491.595726592122
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 1.5387410399488646 + 1.5707963267948966 \\
&= 3.109537366743761
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 1.5387410399488646 + 0.03490658503988659 \\
&= 1.5736476249887512
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (2.3292949232141194, 17.899852673407583) \\
&= 1.4413941943199284
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 1.5736476249887512 - 1.4413941943199284 \\
&= 0.13225343066882278
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.13225343066882278) \\
&= \frac{1}{2} \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 12.2 \cdot 1.0062050722759672 \\
&= 2331.3485527233875
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 2331.3485527233875 \text{ in direction normal to } 1.4413941943199284 \\
&= 2331.3485527233875 \text{ in direction } 3.012190521114825 \\
&= (-2311.8566616284575, 300.8402406087946)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{2331.3485527233875^2}{0.5 \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\
&= 131.65816099321947
\end{aligned}$$

Induced Drag (wing)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 131.65816099321947 \text{ in direction away from } 1.4413941943199284 \\
&= (-16.9893398329662, -130.5573961449771)
\end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (-2311.8566616284575, 300.8402406087946) + (-16.9893398329662, -130.5573961449771) \\
&= (-2328.846001461424, 170.2828444638175)
\end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-2328.846001461424, 170.2828444638175) \text{ in direction } 3.109537366743761 + 90^\circ \\
&= \text{component of } (-2328.846001461424, 170.2828444638175) \text{ in direction } 4.680333693538658 \\
&= -95.55632300410812
\end{aligned}$$

Compute Moment(wing)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= -95.55632300410812 \cdot 0.54 \\
&= -51.60041442221839
\end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 1.5387410399488646 + 3.141592653589793 \\
&= 4.680333693538658
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 1.5387410399488646 + -0.10297442586766545 \\
&= 1.435766614081199
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= 1.435766614081199 - 1.4413941943199284 \\
&= -0.005627580238729335
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(-0.005627580238729335) \\
&= \frac{1}{2} \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 1.97 \cdot -0.03224365965504152 \\
&= -12.063447262790708
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= -12.063447262790708 \text{ in direction normal to } 1.4413941943199284 \\
&= -12.063447262790708 \text{ in direction } 3.012190521114825 \\
&= (11.962587440692863, -1.5566828790444098)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{-12.063447262790708^2}{0.5 \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\ &= 0.04833757574287424 \end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0.04833757574287424 \text{ in direction away from } 1.4413941943199284 \\ &= (-0.006237543459533267, -0.04793343593243174) \end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (11.962587440692863, -1.5566828790444098) + (-0.006237543459533267, -0.04793343593243174) \\ &= (11.956349897233329, -1.6046163149768415) \end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (11.956349897233329, -1.6046163149768415) \text{ in direction } 4.680333693538658 + 90^\circ \\ &= \text{component of } (11.956349897233329, -1.6046163149768415) \text{ in direction } 6.251130020333554 \\ &= 12.001635228733385 \end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= 12.001635228733385 \cdot 4 \\ &= 48.00654091493354 \end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -51.60041442221839 + 48.00654091493354 \\&= -3.5938735072848544\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{-3.5938735072848544}{1000.0} \\&= -0.0035938735072848543\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= 0.1645214403696835 + -0.0035938735072848543 \cdot 0.01 \\&= 0.16448550163461065 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= 1.5387410399488646 + 0.16448550163461065 \cdot 0.01 \\&= 1.5403858949652107\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.27542634472455924) \cdot 1.1657336923917203 \cdot 236.6578434999577^2 \cdot 1.74^4 \\
&= 0.004020010429547757 \cdot 1.1657336923917203 \cdot 236.6578434999577^2 \cdot 1.74^4 \\
&= 2405.8321668332114 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 2405.8321668332114 \text{ Newtons, in direction } 1.5403858949652107 \\
&= (73.15111892945775, 2404.7198025484704)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.1657336923917203 \cdot 18.050771190423074^2 \cdot 0.03 \cdot 1.2 \\
&= 6.8369653080856985 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 6.8369653080856985 \text{ Newtons in direction away from } 1.4413941943199284 \\
&= (-0.8822508697447968, -6.779802948965546)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (73.15111892945775, 2404.7198025484704) + (-2346.424218916462, 170.57317579871815) + \\
&\quad (8.462983261040403, -1.1256964458123806) + (-0.8822508697447968, -6.779802948965546) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (-2265.6923675957087, -3312.6125210475893)
\end{aligned}$$

Linear Acceleration

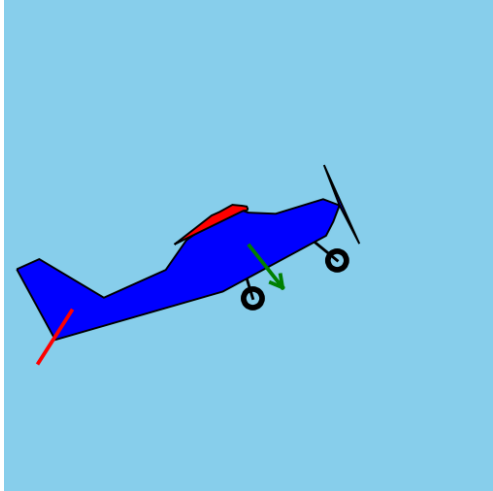
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(-2265.6923675957087, -3312.6125210475893)}{600.0} \\&= (-3.776153945992848, -5.521020868412649)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (2.3292949232141194, 17.899852673407583) + (-3.776153945992848, -5.521020868412649) \cdot 0.01 \\&= (2.291533383754191, 17.844642464723456) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (1797.8474288628177, 514.0760387844257) + (2.291533383754191, 17.844642464723456) \cdot 0.01 \\&= (1797.8703441966552, 514.254485209073)\end{aligned}$$

4 Test 3

```
test_aircraft3 = TecnamP92(angular_velocity=-0.3601050669912345, engine_rpm=5469.66892738982,  
    linear_velocity=(6.700791955156834, -8.46798919852018), pitch_angle_radians=0.42213798000369585,  
    position=(1805.188577984907, 537.0330849941952), stabilizer_angle_radians=0.1937315469713706, throttle=1.0,  
    time_elapsed=62.396)  
simulate_one_time_step(test_aircraft3, 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine_torque}(\text{throttle}, N_e) \\ &= \text{TecnamP92.engine_torque}(1.0, 5469.66892738982) \\ &= 126.2605204690675\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(6.700791955156834, -8.46798919852018)| \\ &= 10.798493130643228\end{aligned}$$

Engine RPM

$$N_e = 5469.66892738982$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{5469.66892738982}{2.43} \\ &= 2250.89256271186 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{10.798493130643228}{\frac{2250.89256271186}{60} \cdot 1.74} \\ &= 0.1654285230044595 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 537.0330849941952 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 537.0330849941952 \\ &= 284.6592849475377 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{284.6592849475377}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 95041.74685991208 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{95041.74685991208}{287.05 \cdot 284.6592849475377} \\ &= 1.163138697009185 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{2250.89256271186 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 235.71291796784942 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{TecnamP92.propeller_torque_coefficient}(0.1654285230044595) \cdot 1.163138697009185 \cdot +235.71291796784942^2 \cdot 1.74^5 \\
&= 0.0002974501956193427 \cdot 1.163138697009185 \cdot +235.71291796784942^2 \cdot 1.74^5 \\
&= 306.5905562306123
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{306.5905562306123}{2.43} \\
&= 126.16895318132194
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= 126.2605204690675 - 126.16895318132194 \\
&= 0.091567287745562
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{0.091567287745562}{0.1} \\
&= 0.91567287745562
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 5469.66892738982 + 0.91567287745562 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 5469.843808027463
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 0.42213798000369585 + 1.5707963267948966 \\
&= 1.9929343067985923
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 0.42213798000369585 + 0.03490658503988659 \\
&= 0.45704456504358243
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (6.700791955156834, -8.46798919852018) \\
&= -0.9013775854875181
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.45704456504358243 - -0.9013775854875181 \\
&= 1.3584221505311005
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(1.3584221505311005) \\
&= \frac{1}{2} \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 12.2 \cdot 0 \\
&= 0.0
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0 \text{ in direction normal to } -0.9013775854875181 \\
&= 0.0 \text{ in direction } 0.6694187413073784 \\
&= (0.0, 0.0)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{0.0^2}{0.5 \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\ &= 0.0 \end{aligned}$$

Induced Drag (wing)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0.0 \text{ in direction away from } -0.9013775854875181 \\ &= (-0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (-0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 1.9929343067985923 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 3.563730633593489 \\ &= -0.0 \end{aligned}$$

Compute Moment(wing)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -0.0 \cdot 0.54 \\ &= -0.0 \end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 0.42213798000369585 + 3.141592653589793 \\
&= 3.563730633593489
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 0.42213798000369585 + 0.1937315469713706 \\
&= 0.6158695269750665
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= 0.6158695269750665 - -0.9013775854875181 \\
&= 1.5172471124625846
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(1.5172471124625846) \\
&= \frac{1}{2} \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 1.97 \cdot 0 \\
&= 0.0
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0 \text{ in direction normal to } -0.9013775854875181 \\
&= 0.0 \text{ in direction } 0.6694187413073784 \\
&= (0.0, 0.0)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{0.0^2}{0.5 \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\ &= 0.0 \end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0.0 \text{ in direction away from } -0.9013775854875181 \\ &= (-0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (-0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 3.563730633593489 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 5.134526960388385 \\ &= 0.0 \end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= 0.0 \cdot 4 \\ &= 0.0 \end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -0.0 + 0.0 \\&= 0.0\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{0.0}{1000.0} \\&= 0.0\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= -0.3601050669912345 + 0.0 \cdot 0.01 \\&= -0.3601050669912345 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= 0.42213798000369585 + -0.3601050669912345 \cdot 0.01 \\&= 0.4185369293337835\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.16542323396027428) \cdot 1.163138697009185 \cdot 235.72045437013296^2 \cdot 1.74^4 \\
&= 0.004230962918447641 \cdot 1.163138697009185 \cdot 235.72045437013296^2 \cdot 1.74^4 \\
&= 2506.4685289805657 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 2506.4685289805657 \text{ Newtons, in direction } 0.4185369293337835 \\
&= (2290.121557932968, 1018.6892247691039)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.163138697009185 \cdot 10.798493130643228^2 \cdot 0.03 \cdot 1.2 \\
&= 2.441351555678485 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 2.441351555678485 \text{ Newtons in direction away from } -0.9013775854875181 \\
&= (-1.514932561986597, 1.9144651344557007)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (2290.121557932968, 1018.6892247691039) + (0.0, 0.0) + \\
&\quad (0.0, 0.0) + (-1.514932561986597, 1.9144651344557007) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (2288.6066253709814, -4859.39631009644)
\end{aligned}$$

Linear Acceleration

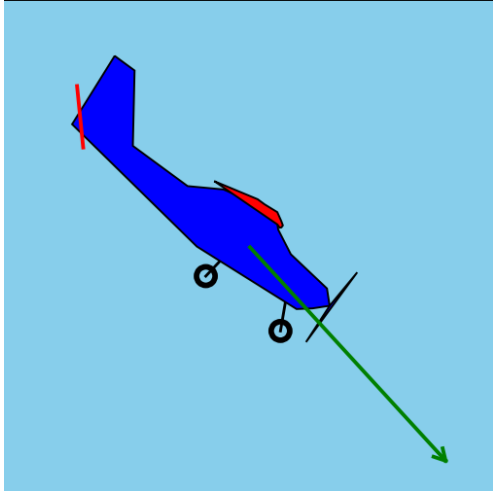
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(2288.6066253709814, -4859.39631009644)}{600.0} \\&= (3.8143443756183024, -8.098993850160735)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (6.700791955156834, -8.46798919852018) + (3.8143443756183024, -8.098993850160735) \cdot 0.01 \\&= (6.738935398913017, -8.548979137021787) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (1805.188577984907, 537.0330849941952) + (6.738935398913017, -8.548979137021787) \cdot 0.01 \\&= (1805.255967338896, 536.9475952028249)\end{aligned}$$

5 Test 4

```
test_aircraft4 = TecnamP92(angular_velocity=2.5462461874185127, engine_rpm=5789.881024080416,  
    linear_velocity=(39.59202073532071, -43.20742419924472), pitch_angle_radians=-0.6335645532198284,  
    position=(1886.9970724837021, 388.531086940641), stabilizer_angle_radians=-0.2792526803190927, throttle=1.0,  
    time_elapsed=67.196)  
simulate_one_time_step(test_aircraft4 , 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine_torque}(\text{throttle}, N_e) \\ &= \text{TecnamP92.engine_torque}(1.0, 5789.881024080416) \\ &= 124.52873489855588\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(39.59202073532071, -43.20742419924472)| \\ &= 58.60383615293066\end{aligned}$$

Engine RPM

$$N_e = 5789.881024080416$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{5789.881024080416}{2.43} \\ &= 2382.6670880989363 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{58.60383615293066}{\frac{2382.6670880989363}{60} \cdot 1.74} \\ &= 0.8481344063556573 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 388.531086940641 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 388.531086940641 \\ &= 285.6245479348858 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{285.6245479348858}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 96746.7345806474 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{96746.7345806474}{287.05 \cdot 285.6245479348858} \\ &= 1.1800033337431777 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{2382.6670880989363 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 249.51231399739342 \end{aligned}$$

Propeller torque load

$$\begin{aligned}Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\&= \text{TecnamP92.propeller_torque_coefficient}(0.8481344063556573) \cdot 1.1800033337431777 \cdot +249.51231399739342^2 \cdot 1.74^5 \\&= 0.0002576820706181458 \cdot 1.1800033337431777 \cdot +249.51231399739342^2 \cdot 1.74^5 \\&= 301.92400595598485\end{aligned}$$

Engine torque load

$$\begin{aligned}Q_e &= \frac{Q_p}{G} \\&= \frac{301.92400595598485}{2.43} \\&= 124.24856212180445\end{aligned}$$

Engine net torque

$$\begin{aligned}\tau_{\text{net}} &= \tau_e - Q_e \\&= 124.52873489855588 - 124.24856212180445 \\&= 0.28017277675142793\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\&= \frac{0.28017277675142793}{0.1} \\&= 2.8017277675142793\end{aligned}$$

Update Engine RPM

$$\begin{aligned}N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\&= 5789.881024080416 + 2.8017277675142793 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\&= 5790.416114668493\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= -0.6335645532198284 + 1.5707963267948966 \\
&= 0.9372317735750681
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= -0.6335645532198284 + 0.03490658503988659 \\
&= -0.5986579681799418
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (39.59202073532071, -43.20742419924472) \\
&= -0.8290350306135054
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= -0.5986579681799418 - -0.8290350306135054 \\
&= 0.2303770624335636
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.2303770624335636) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 12.2 \cdot 1.4559706699252049 \\
&= 35992.9784640703
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 35992.9784640703 \text{ in direction normal to } -0.8290350306135054 \\
&= 35992.9784640703 \text{ in direction } 0.7417612961813912 \\
&= (26536.895718448533, 24316.40730747896)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{35992.9784640703^2}{0.5 \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\
&= 2941.199522358652
\end{aligned}$$

Induced Drag (wing)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 2941.199522358652 \text{ in direction away from } -0.8290350306135054 \\
&= (-1987.037711525575, 2168.4869755887325)
\end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (26536.895718448533, 24316.40730747896) + (-1987.037711525575, 2168.4869755887325) \\
&= (24549.858006922957, 26484.894283067693)
\end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (24549.858006922957, 26484.894283067693) \text{ in direction } 0.9372317735750681 + 90^\circ \\
&= \text{component of } (24549.858006922957, 26484.894283067693) \text{ in direction } 2.5080281003699647 \\
&= -4105.658403789647
\end{aligned}$$

Compute Moment(wing)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= -4105.658403789647 \cdot 0.54 \\
&= -2217.0555380464098
\end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= -0.6335645532198284 + 3.141592653589793 \\
&= 2.5080281003699647
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= -0.6335645532198284 + -0.2792526803190927 \\
&= -0.9128172335389211
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= -0.9128172335389211 - -0.8290350306135054 \\
&= -0.08378220292541572
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(-0.08378220292541572) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 1.97 \cdot -0.480036662593494 \\
&= -1916.2226240569323
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= -1916.2226240569323 \text{ in direction normal to } -0.8290350306135054 \\
&= -1916.2226240569323 \text{ in direction } 0.7417612961813912 \\
&= (-1412.7922199795669, -1294.576103638887)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{-1916.2226240569323^2}{0.5 \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\
&= 114.3113828081542
\end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 114.3113828081542 \text{ in direction away from } -0.8290350306135054 \\
&= (-77.22734441160465, 84.27947267658816)
\end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (-1412.7922199795669, -1294.576103638887) + (-77.22734441160465, 84.27947267658816) \\
&= (-1490.0195643911716, -1210.2966309622989)
\end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-1490.0195643911716, -1210.2966309622989) \text{ in direction } 2.5080281003699647 + 90^\circ \\
&= \text{component of } (-1490.0195643911716, -1210.2966309622989) \text{ in direction } 4.078824427164861 \\
&= 1857.5283636995885
\end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= 1857.5283636995885 \cdot 4 \\
&= 7430.113454798354
\end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -2217.0555380464098 + 7430.113454798354 \\&= 5213.0579167519445\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{5213.0579167519445}{1000.0} \\&= 5.213057916751945\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= 2.5462461874185127 + 5.213057916751945 \cdot 0.01 \\&= 2.598376766586032 \\\theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= -0.6335645532198284 + 2.598376766586032 \cdot 0.01 \\&= -0.6075807855539681\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.8480560305137007) \cdot 1.1800033337431777 \cdot 249.53537348519188^2 \cdot 1.74^4 \\
&= 0.0012214842156256516 \cdot 1.1800033337431777 \cdot 249.53537348519188^2 \cdot 1.74^4 \\
&= 822.6826232888254 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 822.6826232888254 \text{ Newtons, in direction } -0.6075807855539681 \\
&= (675.4483541415901, -469.6554264019531)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 0.03 \cdot 1.2 \\
&= 72.94706624538492 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 72.94706624538492 \text{ Newtons in direction away from } -0.8290350306135054 \\
&= (-49.28212808170688, 53.78239791555239)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (675.4483541415901, -469.6554264019531) + (26382.237877606363, 28843.298893073534) + \\
&\quad (-1011.3899260294866, -852.9733499106427) + (-49.28212808170688, 53.78239791555239) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (25997.01417763676, 21694.452514676486)
\end{aligned}$$

Linear Acceleration

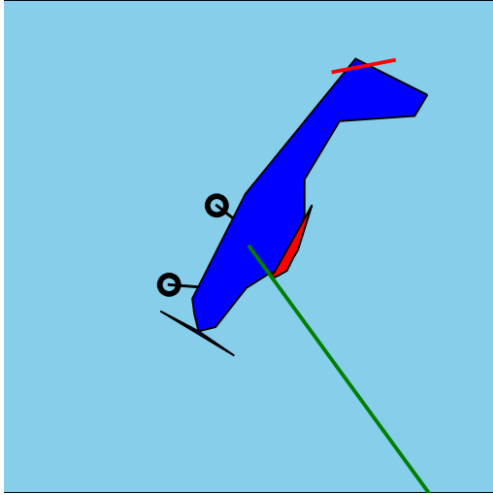
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(25997.01417763676, 21694.452514676486)}{600.0} \\&= (43.32835696272794, 36.15742085779414)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (39.59202073532071, -43.20742419924472) + (43.32835696272794, 36.15742085779414) \cdot 0.01 \\&= (40.02530430494799, -42.84584999066678) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (1886.9970724837021, 388.531086940641) + (40.02530430494799, -42.84584999066678) \cdot 0.01 \\&= (1887.3973255267517, 388.10262844073435)\end{aligned}$$

6 Test 5

```
test_aircraft5 = TecnamP92(angular_velocity=-1.4479315865978675, engine_rpm=5800, linear_velocity=(40.81904228175883,  
-56.020830283261), pitch_angle_radians=-2.112939056059267, position=(1950.0811530507483, 316.9117787679673),  
stabilizer_angle_radians=-0.2792526803190927, throttle=1.0, time_elapsed=68.696)  
simulate_one_time_step(test_aircraft5, 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine_torque}(\textit{throttle}, N_e) \\ &= \text{TecnamP92.engine_torque}(1.0, 5800) \\ &= 124.45756\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(40.81904228175883, -56.020830283261)| \\ &= 69.31470001685031\end{aligned}$$

Engine RPM

$$N_e = 5800$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{5800}{2.43} \\ &= 2386.8312757201643 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{69.31470001685031}{\frac{2386.8312757201643}{60} \cdot 1.74} \\ &= 1.0013954877583013 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 316.9117787679673 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 316.9117787679673 \\ &= 286.0900734380082 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{286.0900734380082}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 97577.81771810376 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{97577.81771810376}{287.05 \cdot 286.0900734380082} \\ &= 1.1882033177099898 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{2386.8312757201643 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 249.94838670536075 \end{aligned}$$

Propeller torque load

$$\begin{aligned}Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\&= \text{TecnamP92.propeller_torque_coefficient}(1.0013954877583013) \cdot 1.1882033177099898 \cdot +249.94838670536075^2 \cdot 1.74^5 \\&= 0.00024139121132743182 \cdot 1.1882033177099898 \cdot +249.94838670536075^2 \cdot 1.74^5 \\&= 285.797964959147\end{aligned}$$

Engine torque load

$$\begin{aligned}Q_e &= \frac{Q_p}{G} \\&= \frac{285.797964959147}{2.43} \\&= 117.61233125890823\end{aligned}$$

Engine net torque

$$\begin{aligned}\tau_{\text{net}} &= \tau_e - Q_e \\&= 124.45756 - 117.61233125890823 \\&= 6.845228741091773\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\&= \frac{6.845228741091773}{0.1} \\&= 68.45228741091773\end{aligned}$$

Update Engine RPM

$$\begin{aligned}N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\&= 5800 + 68.45228741091773 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\&= 5813.073423888874 \\&= 5800 \text{ reduced to max TecnamP92 engine RPM}\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= -2.112939056059267 + 1.5707963267948966 \\
&= -0.5421427292643703
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= -2.112939056059267 + 0.03490658503988659 \\
&= -2.0780324710193803
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (40.81904228175883, -56.020830283261) \\
&= -0.9411060658643418
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= -2.0780324710193803 - -0.9411060658643418 \\
&= -1.1369264051550385
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(-1.1369264051550385) \\
&= \frac{1}{2} \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 12.2 \cdot 0 \\
&= 0.0
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0 \text{ in direction normal to } -0.9411060658643418 \\
&= 0.0 \text{ in direction } 0.6296902609305548 \\
&= (0.0, 0.0)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{0.0^2}{0.5 \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\ &= 0.0 \end{aligned}$$

Induced Drag (wing)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0.0 \text{ in direction away from } -0.9411060658643418 \\ &= (-0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (-0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } -0.5421427292643703 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 1.0286535975305262 \\ &= 0.0 \end{aligned}$$

Compute Moment(wing)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= 0.0 \cdot 0.54 \\ &= 0.0 \end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= -2.112939056059267 + 3.141592653589793 \\
&= 1.0286535975305262
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= -2.112939056059267 + -0.2792526803190927 \\
&= -2.3921917363783596
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= -2.3921917363783596 - -0.9411060658643418 \\
&= -1.4510856705140178
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(-1.4510856705140178) \\
&= \frac{1}{2} \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 1.97 \cdot 0 \\
&= 0.0
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0 \text{ in direction normal to } -0.9411060658643418 \\
&= 0.0 \text{ in direction } 0.6296902609305548 \\
&= (0.0, 0.0)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{0.0^2}{0.5 \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\ &= 0.0 \end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0.0 \text{ in direction away from } -0.9411060658643418 \\ &= (-0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (-0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 1.0286535975305262 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 2.5994499243254228 \\ &= -0.0 \end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -0.0 \cdot 4 \\ &= -0.0 \end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= 0.0 + -0.0 \\&= 0.0\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{0.0}{1000.0} \\&= 0.0\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= -1.4479315865978675 + 0.0 \cdot 0.01 \\&= -1.4479315865978675\end{aligned}$$

$$\begin{aligned}\theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= -2.112939056059267 + -1.4479315865978675 \cdot 0.01 \\&= -2.1274183719252457\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(1.0013954877583013) \cdot 1.1882033177099898 \cdot 249.94838670536075^2 \cdot 1.74^4 \\
&= -1.2149214626684608e - 05 \cdot 1.1882033177099898 \cdot 249.94838670536075^2 \cdot 1.74^4 \\
&= -8.266784833280058 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= -8.266784833280058 \text{ Newtons, in direction } -2.1274183719252457 \\
&= (4.3675175449358425, 7.018868995388558)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.1882033177099898 \cdot 69.31470001685031^2 \cdot 0.03 \cdot 1.2 \\
&= 102.75760224012697 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 102.75760224012697 \text{ Newtons in direction away from } -0.9411060658643418 \\
&= (-60.51338185972428, 83.04971664033141)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (4.3675175449358425, 7.018868995388558) + (0.0, 0.0) + \\
&\quad (0.0, 0.0) + (-60.51338185972428, 83.04971664033141) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (-56.14586431478843, -5789.93141436428)
\end{aligned}$$

Linear Acceleration

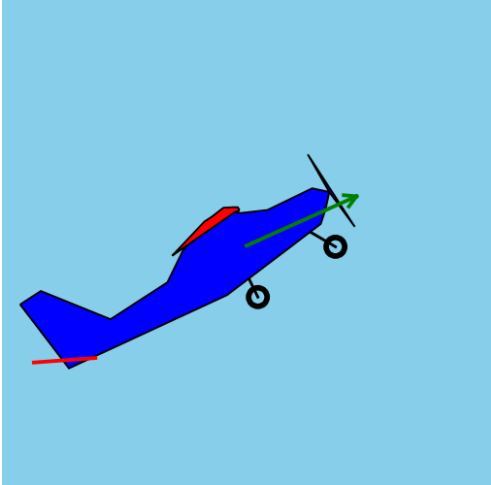
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(-56.14586431478843, -5789.93141436428)}{600.0} \\&= (-0.09357644052464739, -9.649885690607134)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (40.81904228175883, -56.020830283261) + (-0.09357644052464739, -9.649885690607134) \cdot 0.01 \\&= (40.818106517353584, -56.11732914016707) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (1950.0811530507483, 316.9117787679673) + (40.818106517353584, -56.11732914016707) \cdot 0.01 \\&= (1950.4893341159218, 316.35060547656565)\end{aligned}$$

7 Test 6

```
test_aircraft6 = TecnamP92(angular_velocity=0.03397972101623799, engine_rpm=4875.897336540748,  
    linear_velocity=(22.2882025836437, 10.012123108247119), pitch_angle_radians=0.5759732753508979,  
    position=(256.7063983204896, 30.26187784869959), stabilizer_angle_radians=-0.16755160819145562, throttle=0.777,  
    time_elapsed=14.697)  
simulate_one_time_step(test_aircraft6 , 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine.torque}(\text{throttle}, N_e) \\ &= \text{TecnamP92.engine.torque}(0.777, 4875.897336540748) \\ &= 102.20669446852743\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(22.2882025836437, 10.012123108247119)| \\ &= 24.433718168634048\end{aligned}$$

Engine RPM

$$N_e = 4875.897336540748$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{4875.897336540748}{2.43} \\ &= 2006.5421138027768 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{24.433718168634048}{\frac{2006.5421138027768}{60} \cdot 1.74} \\ &= 0.4198974943108424 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 30.26187784869959 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 30.26187784869959 \\ &= 287.95329779398344 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{287.95329779398344}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 100962.22942392105 \\ \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{100962.22942392105}{287.05 \cdot 287.95329779398344} \\ &= 1.2214602101665812 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{2006.5421138027768 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 210.1245987947113 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{TecnamP92.propeller_torque_coefficient}(0.4198974943108424) \cdot 1.2214602101665812 \cdot +210.1245987947113^2 \cdot 1.74^5 \\
&= 0.0002888900054179009 \cdot 1.2214602101665812 \cdot +210.1245987947113^2 \cdot 1.74^5 \\
&= 248.49169416241025
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{248.49169416241025}{2.43} \\
&= 102.25995644543632
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= 102.20669446852743 - 102.25995644543632 \\
&= -0.053261976908885345
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{-0.053261976908885345}{0.1} \\
&= -0.5326197690888534
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 4875.897336540748 + -0.5326197690888534 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 4875.795613657901
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 0.5759732753508979 + 1.5707963267948966 \\
&= 2.1467696021457945
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 0.5759732753508979 + 0.03490658503988659 \\
&= 0.6108798603907846
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (22.2882025836437, 10.012123108247119) \\
&= 0.42219824664764904
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.6108798603907846 - 0.42219824664764904 \\
&= 0.18868161374313552
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.18868161374313552) \\
&= \frac{1}{2} \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 12.2 \cdot 1.2648528111359405 \\
&= 5626.369754941989
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 5626.369754941989 \text{ in direction normal to } 0.42219824664764904 \\
&= 5626.369754941989 \text{ in direction } 1.9929945734425456 \\
&= (-2305.4987476818624, 5132.320346954518)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{5626.369754941989^2}{0.5 \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\ &= 399.41312685555147 \end{aligned}$$

Induced Drag (wing)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 399.41312685555147 \text{ in direction away from } 0.42219824664764904 \\ &= (-364.340810697858, -163.66618332617304) \end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (-2305.4987476818624, 5132.320346954518) + (-364.340810697858, -163.66618332617304) \\ &= (-2669.8395583797205, 4968.654163628345) \end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (-2669.8395583797205, 4968.654163628345) \text{ in direction } 2.1467696021457945 + 90^\circ \\ &= \text{component of } (-2669.8395583797205, 4968.654163628345) \text{ in direction } 3.717565928940691 \\ &= -467.08934392931036 \end{aligned}$$

Compute Moment(wing)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -467.08934392931036 \cdot 0.54 \\ &= -252.22824572182762 \end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 0.5759732753508979 + 3.141592653589793 \\
&= 3.717565928940691
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 0.5759732753508979 + -0.16755160819145562 \\
&= 0.4084216671594423
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= 0.4084216671594423 - 0.42219824664764904 \\
&= -0.013776579488206764
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(-0.013776579488206764) \\
&= \frac{1}{2} \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 1.97 \cdot -0.07893398608007474 \\
&= -56.696821149154374
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= -56.696821149154374 \text{ in direction normal to } 0.42219824664764904 \\
&= -56.696821149154374 \text{ in direction } 1.9929945734425456 \\
&= (23.232467088055017, -51.718294649201646)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{-56.696821149154374^2}{0.5 \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\
&= 0.5561498850548799
\end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 0.5561498850548799 \text{ in direction away from } 0.42219824664764904 \\
&= (-0.5073145732230687, -0.22789168137966748)
\end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (23.232467088055017, -51.718294649201646) + (-0.5073145732230687, -0.22789168137966748) \\
&= (22.725152514831947, -51.94618633058131)
\end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (22.725152514831947, -51.94618633058131) \text{ in direction } 3.717565928940691 + 90^\circ \\
&= \text{component of } (22.725152514831947, -51.94618633058131) \text{ in direction } 5.288362255735588 \\
&= 55.942607715966645
\end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= 55.942607715966645 \cdot 4 \\
&= 223.77043086386658
\end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -252.22824572182762 + 223.77043086386658 \\&= -28.45781485796104\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{-28.45781485796104}{1000.0} \\&= -0.02845781485796104\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= 0.03397972101623799 + -0.02845781485796104 \cdot 0.01 \\&= 0.03369514286765838 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= 0.5759732753508979 + 0.03369514286765838 \cdot 0.01 \\&= 0.5763102267795746\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.4199062545598368) \cdot 1.2214602101665812 \cdot 210.12021509290807^2 \cdot 1.74^4 \\
&= 0.0035830025076096536 \cdot 1.2214602101665812 \cdot 210.12021509290807^2 \cdot 1.74^4 \\
&= 1771.1653313622721 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 1771.1653313622721 \text{ Newtons, in direction } 0.5763102267795746 \\
&= (1485.0849985744267, 965.1679532748809)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.2214602101665812 \cdot 24.433718168634048^2 \cdot 0.03 \cdot 1.2 \\
&= 13.125956166121984 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 13.125956166121984 \text{ Newtons in direction away from } 0.42219824664764904 \\
&= (-11.973370901450036, -5.37857922980273)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (1485.0849985744267, 965.1679532748809) + (-2673.54504286219, 4974.521137760823) + \\
&\quad (22.181438950063654, -50.67023448576707) + (-11.973370901450036, -5.37857922980273) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (-1178.2519762391494, 3.6402773201343734)
\end{aligned}$$

Linear Acceleration

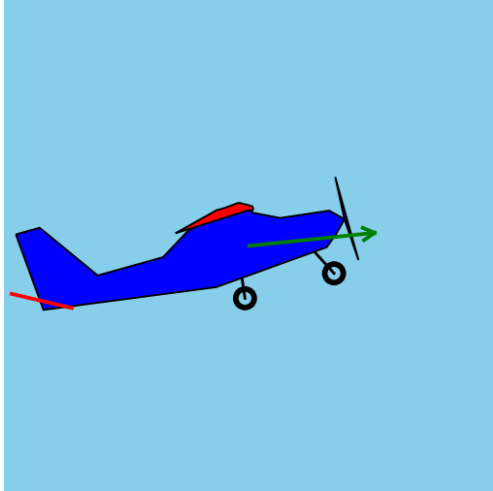
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(-1178.2519762391494, 3.6402773201343734)}{600.0} \\&= (-1.9637532937319158, 0.006067128866890622)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (22.2882025836437, 10.012123108247119) + (-1.9637532937319158, 0.006067128866890622) \cdot 0.01 \\&= (22.26856505070638, 10.012183779535787) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (256.7063983204896, 30.26187784869959) + (22.26856505070638, 10.012183779535787) \cdot 0.01 \\&= (256.92908397099666, 30.361999686494947)\end{aligned}$$

8 Test 7

```
test_aircraft7 = TecnamP92(angular_velocity=-0.01490415703929847, engine_rpm=3601.7943442586134,  
    linear_velocity=(25.348417258520143, 2.624451225922511), pitch_angle_radians=0.2730575554235457,  
    position=(842.996063815359, 3.4635132479641855), stabilizer_angle_radians=-0.16755160819145562,  
    throttle=0.38400000000000006, time_elapsed=39.597)  
simulate_one_time_step(test_aircraft7, 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{TecnamP92.engine_torque}(\text{throttle}, N_e) \\ &= \text{TecnamP92.engine_torque}(0.38400000000000006, 3601.7943442586134) \\ &= 53.96503823035123\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(25.348417258520143, 2.624451225922511)| \\ &= 25.48391653080994\end{aligned}$$

Engine RPM

$$N_e = 3601.7943442586134$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{3601.7943442586134}{2.43} \\ &= 1482.219894756631 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{25.48391653080994}{\frac{1482.219894756631}{60} \cdot 1.74} \\ &= 0.5928646252491483 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 3.4635132479641855 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 3.4635132479641855 \\ &= 288.1274871638882 \\ P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{288.1274871638882}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 101283.4270136369 \\ \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{101283.4270136369}{287.05 \cdot 288.1274871638882} \\ &= 1.2246053284910092 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{1482.219894756631 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 155.21770441240227 \end{aligned}$$

Propeller torque load

$$\begin{aligned}Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\&= \text{TecnamP92.propeller_torque_coefficient}(0.5928646252491483) \cdot 1.2246053284910092 \cdot +155.21770441240227^2 \cdot 1.74^5 \\&= 0.00027882250207633257 \cdot 1.2246053284910092 \cdot +155.21770441240227^2 \cdot 1.74^5 \\&= 131.20570484957236\end{aligned}$$

Engine torque load

$$\begin{aligned}Q_e &= \frac{Q_p}{G} \\&= \frac{131.20570484957236}{2.43} \\&= 53.994117222046235\end{aligned}$$

Engine net torque

$$\begin{aligned}\tau_{\text{net}} &= \tau_e - Q_e \\&= 53.96503823035123 - 53.994117222046235 \\&= -0.029078991695001832\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\&= \frac{-0.029078991695001832}{0.1} \\&= -0.2907899169500183\end{aligned}$$

Update Engine RPM

$$\begin{aligned}N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\&= 3601.7943442586134 + -0.2907899169500183 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\&= 3601.7388074753926\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 0.2730575554235457 + 1.5707963267948966 \\
&= 1.8438538822184423
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 0.2730575554235457 + 0.03490658503988659 \\
&= 0.3079641404634323
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (25.348417258520143, 2.624451225922511) \\
&= 0.10316752469105317
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.3079641404634323 - 0.10316752469105317 \\
&= 0.20479661577237912
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.20479661577237912) \\
&= \frac{1}{2} \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 12.2 \cdot 1.3387185393855738 \\
&= 6494.528178979069
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 6494.528178979069 \text{ in direction normal to } 0.10316752469105317 \\
&= 6494.528178979069 \text{ in direction } 1.6739638514859498 \\
&= (-668.8364569277688, 6459.9964443827275)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{6494.528178979069^2}{0.5 \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\
&= 487.96756678290427
\end{aligned}$$

Induced Drag (wing)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 487.96756678290427 \text{ in direction away from } 0.10316752469105317 \\
&= (-485.37301856578983, -50.253149954620255)
\end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (-668.8364569277688, 6459.9964443827275) + (-485.37301856578983, -50.253149954620255) \\
&= (-1154.2094754935588, 6409.743294428107)
\end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-1154.2094754935588, 6409.743294428107) \text{ in direction } 1.8438538822184423 + 90^\circ \\
&= \text{component of } (-1154.2094754935588, 6409.743294428107) \text{ in direction } 3.414650209013339 \\
&= -617.1131473635216
\end{aligned}$$

Compute Moment(wing)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= -617.1131473635216 \cdot 0.54 \\
&= -333.24109957630174
\end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 0.2730575554235457 + 3.141592653589793 \\
&= 3.414650209013339
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 0.2730575554235457 + -0.16755160819145562 \\
&= 0.10550594723209009
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= 0.10550594723209009 - 0.10316752469105317 \\
&= 0.0023384225410369164
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(0.0023384225410369164) \\
&= \frac{1}{2} \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 1.97 \cdot 0.013398174231967286 \\
&= 10.49567427618037
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 10.49567427618037 \text{ in direction normal to } 0.10316752469105317 \\
&= 10.49567427618037 \text{ in direction } 1.6739638514859498 \\
&= (-1.0808929305549524, 10.439868245545567)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{10.49567427618037^2}{0.5 \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\
&= 0.017475317409551144
\end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 0.017475317409551144 \text{ in direction away from } 0.10316752469105317 \\
&= (-0.017382400263587155, -0.0017996887622604319)
\end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (-1.0808929305549524, 10.439868245545567) + (-0.017382400263587155, -0.0017996887622604319) \\
&= (-1.0982753308185396, 10.438068556783307)
\end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-1.0982753308185396, 10.438068556783307) \text{ in direction } 3.414650209013339 + 90^\circ \\
&= \text{component of } (-1.0982753308185396, 10.438068556783307) \text{ in direction } 4.985446535808236 \\
&= -10.34752651422579
\end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= -10.34752651422579 \cdot 4 \\
&= -41.39010605690316
\end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -333.24109957630174 + -41.39010605690316 \\&= -374.6312056332049\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{-374.6312056332049}{1000.0} \\&= -0.3746312056332049\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= -0.01490415703929847 + -0.3746312056332049 \cdot 0.01 \\&= -0.01865046909563052 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= 0.2730575554235457 + -0.01865046909563052 \cdot 0.01 \\&= 0.2728710507325894\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.5928737668876546) \cdot 1.2246053284910092 \cdot 155.21531107975247^2 \cdot 1.74^4 \\
&= 0.0028209780299335275 \cdot 1.2246053284910092 \cdot 155.21531107975247^2 \cdot 1.74^4 \\
&= 762.8900682751506 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 762.8900682751506 \text{ Newtons, in direction } 0.2728710507325894 \\
&= (734.6639945990516, 205.5968660087713)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.2246053284910092 \cdot 25.48391653080994^2 \cdot 0.03 \cdot 1.2 \\
&= 14.315317931233865 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 14.315317931233865 \text{ Newtons in direction away from } 0.10316752469105317 \\
&= (-14.239202662219617, -1.474257445815891)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (734.6639945990516, 205.5968660087713) + (-1153.1626741610717, 6405.682251397015) + \\
&\quad (-1.0094048064535612, 9.60569552062225) + (-14.239202662219617, -1.474257445815891) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (-433.7472870306932, 739.4105554805919)
\end{aligned}$$

Linear Acceleration

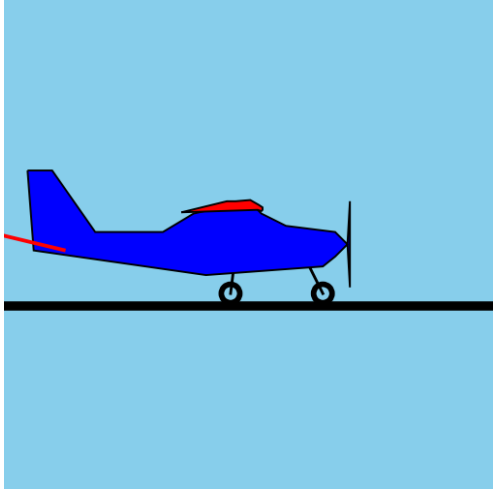
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(-433.7472870306932, 739.4105554805919)}{600.0} \\&= (-0.7229121450511554, 1.2323509258009866)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (25.348417258520143, 2.624451225922511) + (-0.7229121450511554, 1.2323509258009866) \cdot 0.01 \\&= (25.341188137069633, 2.636774735180521) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (842.996063815359, 3.4635132479641855) + (25.341188137069633, 2.636774735180521) \cdot 0.01 \\&= (843.2494756967296, 3.489880995315991)\end{aligned}$$

9 Test 8

```
test_aircraft8 = TecnamP92(angular_velocity=2.316797479594637e-08, engine_rpm=1274.5967848636499,  
    pitch_angle_radians=5.3949531555335793e-08, position=(0.004002432380726907, 0),  
    stabilizer_angle_radians=-0.07853981633974483, time_elapsed=5.099)  
simulate_one_time_step(test_aircraft8, 0.01)
```



Engine torque generated

```
 $\tau_e = \text{TecnamP92.engine.torque}(\textit{throttle}, N_e)$   
= TecnamP92.engine.torque(0, 1274.5967848636499)  
= 7.257837429312109
```

Airspeed

```
 $|v| = |(0, 0)|$   
= 0.0
```


Engine RPM

$$N_e = 1274.5967848636499$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{1274.5967848636499}{2.43} \\ &= 524.5254258698147 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{0.0}{\frac{524.5254258698147}{60} \cdot 1.74} \\ &= 0.0 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 0 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 0 \\ &= 288.15 \\ P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{288.15}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 101325.0 \\ \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{101325.0}{287.05 \cdot 288.15} \\ &= 1.2250122659906946 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{524.5254258698147 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 54.92817415112225 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{TecnamP92.propeller_torque_coefficient}(0.0) \cdot 1.2250122659906946 \cdot +54.92817415112225^2 \cdot 1.74^5 \\
&= 0.0002990229885057471 \cdot 1.2250122659906946 \cdot +54.92817415112225^2 \cdot 1.74^5 \\
&= 17.62712898233998
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{17.62712898233998}{2.43} \\
&= 7.253962544172831
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= 7.257837429312109 - 7.253962544172831 \\
&= 0.003874885139278028
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{0.003874885139278028}{0.1} \\
&= 0.03874885139278028
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 1274.5967848636499 + 0.03874885139278028 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 1274.6041853491358
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 5.3949531555335793e - 08 + 1.5707963267948966 \\
&= 1.570796380744428
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 5.3949531555335793e - 08 + 0.03490658503988659 \\
&= 0.03490663898941815
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (0, 0) \\
&= 0.0
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.03490663898941815 - 0.0 \\
&= 0.03490663898941815
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.03490663898941815) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 12.2 \cdot 0.5600002472864372 \\
&= 0.0
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 0.0 \text{ in direction normal to } 0.0 \\
&= 0.0 \text{ in direction } 1.5707963267948966 \\
&= (0.0, 0.0)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$|F_i| = 0 \text{ since aircraft is stationary}$$

Induced Drag (wing)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0 \text{ in direction away from } 0.0 \\ &= (0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (0.0, 0.0) + (0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 1.570796380744428 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 3.1415927075393246 \\ &= -0.0 \end{aligned}$$

Compute Moment(wing)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -0.0 \cdot 0.54 \\ &= -0.0 \end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 5.3949531555335793e - 08 + 3.141592653589793 \\
&= 3.1415927075393246
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 5.3949531555335793e - 08 + -0.07853981633974483 \\
&= -0.07853976239021328
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= -0.07853976239021328 - 0.0 \\
&= -0.07853976239021328
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(-0.07853976239021328) \\
&= \frac{1}{2} \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 1.97 \cdot -0.4499996908919536 \\
&= -0.0
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= -0.0 \text{ in direction normal to } 0.0 \\
&= -0.0 \text{ in direction } 1.5707963267948966 \\
&= (-0.0, -0.0)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$|F_i| = 0 \text{ since aircraft is stationary}$$

Induced Drag (stabilizer)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0 \text{ in direction away from } 0.0 \\ &= (0.0, 0.0) \end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (-0.0, -0.0) + (0.0, 0.0) \\ &= (0.0, 0.0) \end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 3.1415927075393246 + 90^\circ \\ &= \text{component of } (0.0, 0.0) \text{ in direction } 4.712389034334221 \\ &= 0.0 \end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= 0.0 \cdot 4 \\ &= 0.0 \end{aligned}$$

Net Moment

$$\begin{aligned} M_{net} &= M_w + M_s \\ &= -0.0 + 0.0 \\ &= 0.0 \end{aligned}$$

Angular Acceleration

$$\begin{aligned} \alpha_t &= \frac{M_{net}}{I_p} \\ &= \frac{0.0}{1000.0} \\ &= 0.0 \end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned} \omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\ &= 2.316797479594637e-08 + 0.0 \cdot 0.01 \\ &= 2.316797479594637e-08 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\ &= 5.3949531555335793e-08 + 2.316797479594637e-08 \cdot 0.01 \\ &= 5.418121130329526e-08 \end{aligned}$$

Engine Thrust Force

$$\begin{aligned}|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\&= \text{TecnamP92.propeller_thrust_coefficient}(0.0) \cdot 1.2250122659906946 \cdot 54.928493071709845^2 \cdot 1.74^4 \\&= 0.00435 \cdot 1.2250122659906946 \cdot 54.928493071709845^2 \cdot 1.74^4 \\&= 147.37440223346297 \\F_t &= |F_t| \text{ in direction } \theta \\&= 147.37440223346297 \text{ Newtons, in direction } 5.418121130329526e - 08 \\&= (147.37440223346277, 7.984923628108082e - 06)\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\&= 0.5 \cdot 1.2250122659906946 \cdot 0.0^2 \cdot 0.03 \cdot 1.2 \\&= 0.0 \\F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\&= 0.0 \text{ Newtons in direction away from } 0.0 \\&= (-0.0, -0.0)\end{aligned}$$

Weight Force

$$\begin{aligned}F_g &= (0, -m \cdot g) \\&= (0, -600.0 \cdot 9.8) \\&= (0, -5880.0)\end{aligned}$$

Rolling Resistance Force

$$\begin{aligned}|F_r| &\leq \mu_r \cdot |F_g| \\&= 0.04 \cdot -5880.0 \\&= -235.20000000000002 \\F_r &= (0, 0) \text{ since not on ground or not moving}\end{aligned}$$

Net Force

$$\begin{aligned}F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\&= (147.37440223346277, 7.984923628108082e - 06) + (0.0, 0.0) + \\&\quad (0.0, 0.0) + (-0.0, -0.0) + \\&\quad (0, -5880.0) + (0, 0) \\&= (147.37440223346277, -5879.999992015077)\end{aligned}$$

Linear Acceleration

$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(147.37440223346277, -5879.999992015077)}{600.0} \\&= (0.24562400372243798, -9.799999986691795)\end{aligned}$$

Update Linear Velocity and Position

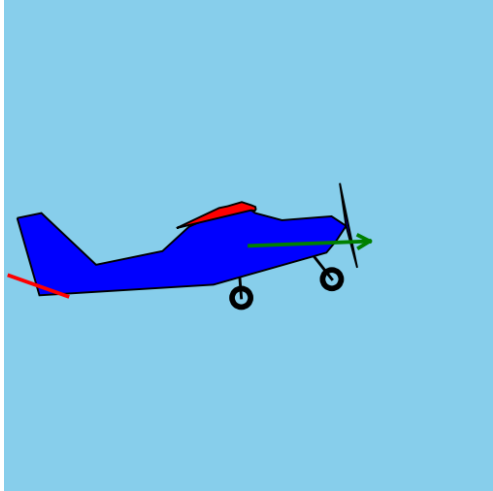
$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (0, 0) + (0.24562400372243798, -9.799999986691795) \cdot 0.01 \\&= (0.00245624003722438, -0.09799999986691794) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (0.004002432380726907, 0) + (0.00245624003722438, -0.09799999986691794) \cdot 0.01 \\&= (0.0040269947810991505, -0.0009799999986691795)\end{aligned}$$

Correct for on ground

position corrected from $(0.0040269947810991505, -0.0009799999986691795)$ to $(0.0040269947810991505, 0)$
linear velocity corrected from $(0.00245624003722438, -0.09799999986691794)$ to $(0.00245624003722438, 0)$

10 Test 9

```
test_aircraft9 = TecnamP92(angular_velocity=-0.008569412357370453, engine_rpm=2913.6305800661235,  
    linear_velocity=(24.396185426677608, 0.9305979895790267), pitch_angle_radians=0.20287459547633346,  
    position=(1912.6724958372301, 180.34367644747098), stabilizer_angle_radians=-0.18151424220741028, throttle=0.223,  
    time_elapsed=83.997)  
simulate_one_time_step(test_aircraft9 , 0.01)
```



Engine torque generated

```
 $\tau_e$  = TecnamP92.engine_torque(throttle,  $N_e$ )  
      = TecnamP92.engine_torque(0.223, 2913.6305800661235)  
      = 33.6843328954465
```

Airspeed

```
|v| = |(24.396185426677608, 0.9305979895790267)|  
      = 24.413927909925626
```

Engine RPM

$$N_e = 2913.6305800661235$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{2913.6305800661235}{2.43} \\ &= 1199.0249300683636 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{24.413927909925626}{\frac{1199.0249300683636}{60} \cdot 1.74} \\ &= 0.7021201661360653 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 180.34367644747098 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 180.34367644747098 \\ &= 286.9777661030914 \end{aligned}$$

$$\begin{aligned} P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{286.9777661030914}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 99178.59689922619 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{99178.59689922619}{287.05 \cdot 286.9777661030914} \\ &= 1.2039602767160136 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{1199.0249300683636 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 125.56159705912621 \end{aligned}$$

Propeller torque load

$$\begin{aligned}Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\&= \text{TecnamP92.propeller_torque_coefficient}(0.7021201661360653) \cdot 1.2039602767160136 \cdot +125.56159705912621^2 \cdot 1.74^5 \\&= 0.0002706912225462681 \cdot 1.2039602767160136 \cdot +125.56159705912621^2 \cdot 1.74^5 \\&= 81.94948620758665\end{aligned}$$

Engine torque load

$$\begin{aligned}Q_e &= \frac{Q_p}{G} \\&= \frac{81.94948620758665}{2.43} \\&= 33.7240683981838\end{aligned}$$

Engine net torque

$$\begin{aligned}\tau_{\text{net}} &= \tau_e - Q_e \\&= 33.6843328954465 - 33.7240683981838 \\&= -0.039735502737300976\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\&= \frac{-0.039735502737300976}{0.1} \\&= -0.39735502737300976\end{aligned}$$

Update Engine RPM

$$\begin{aligned}N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\&= 2913.6305800661235 + -0.39735502737300976 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\&= 2913.554690846001\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= 0.20287459547633346 + 1.5707963267948966 \\
&= 1.7736709222712301
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= 0.20287459547633346 + 0.03490658503988659 \\
&= 0.23778118051622005
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (24.396185426677608, 0.9305979895790267) \\
&= 0.038126740261677436
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= 0.23778118051622005 - 0.038126740261677436 \\
&= 0.1996544402545426
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 12.2 \cdot \text{TecnamP92.wing_lift_coefficient}(0.1996544402545426) \\
&= \frac{1}{2} \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 12.2 \cdot 1.3151485430105714 \\
&= 5756.94548522167
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 5756.94548522167 \text{ in direction normal to } 0.038126740261677436 \\
&= 5756.94548522167 \text{ in direction } 1.608923067056574 \\
&= (-219.44039133847212, 5752.761704995557)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{5756.94548522167^2}{0.5 \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 12.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\
&= 424.9335279752917
\end{aligned}$$

Induced Drag (wing)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 424.9335279752917 \text{ in direction away from } 0.038126740261677436 \\
&= (-424.6247134318295, -16.197405362115457)
\end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (-219.44039133847212, 5752.761704995557) + (-424.6247134318295, -16.197405362115457) \\
&= (-644.0651047703016, 5736.564299633442)
\end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-644.0651047703016, 5736.564299633442) \text{ in direction } 1.7736709222712301 + 90^\circ \\
&= \text{component of } (-644.0651047703016, 5736.564299633442) \text{ in direction } 3.3444672490661267 \\
&= -524.9799991496093
\end{aligned}$$

Compute Moment(wing)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= -524.9799991496093 \cdot 0.54 \\
&= -283.48919954078906
\end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= 0.20287459547633346 + 3.141592653589793 \\
&= 3.3444672490661267
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= 0.20287459547633346 + -0.18151424220741028 \\
&= 0.021360353268923188
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= 0.021360353268923188 - 0.038126740261677436 \\
&= -0.016766386992754248
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 1.97 \cdot \text{TecnamP92.stabilizer_lift_coefficient}(-0.016766386992754248) \\
&= \frac{1}{2} \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 1.97 \cdot -0.09606432123678588 \\
&= -67.90250915980108
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= -67.90250915980108 \text{ in direction normal to } 0.038126740261677436 \\
&= -67.90250915980108 \text{ in direction } 1.608923067056574 \\
&= (2.5882741500924937, -67.85316195374268)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{-67.90250915980108^2}{0.5 \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 1.97 \cdot 3.141592653589793 \cdot +0.6 \cdot AR} \\ &= 0.8106195033634344 \end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 0.8106195033634344 \text{ in direction away from } 0.038126740261677436 \\ &= (-0.8100303968906043, -0.030898791989832627) \end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (2.5882741500924937, -67.85316195374268) + (-0.8100303968906043, -0.030898791989832627) \\ &= (1.7782437532018895, -67.88406074573251) \end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (1.7782437532018895, -67.88406074573251) \text{ in direction } 3.3444672490661267 + 90^\circ \\ &= \text{component of } (1.7782437532018895, -67.88406074573251) \text{ in direction } 4.915263575861023 \\ &= 66.85014696626193 \end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= 66.85014696626193 \cdot 4 \\ &= 267.4005878650477 \end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -283.48919954078906 + 267.4005878650477 \\&= -16.088611675741333\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{-16.088611675741333}{1000.0} \\&= -0.016088611675741334\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= -0.008569412357370453 + -0.016088611675741334 \cdot 0.01 \\&= -0.008730298474127866 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= 0.20287459547633346 + -0.008730298474127866 \cdot 0.01 \\&= 0.2027872924915922\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{TecnamP92.propeller_thrust_coefficient}(0.7021384542265575) \cdot 1.2039602767160136 \cdot 125.55832664737831^2 \cdot 1.74^4 \\
&= 0.0022054569212690807 \cdot 1.2039602767160136 \cdot 125.55832664737831^2 \cdot 1.74^4 \\
&= 383.7061575722231 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 383.7061575722231 \text{ Newtons, in direction } 0.2027872924915922 \\
&= (375.84364312296117, 77.27853054309396)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.2039602767160136 \cdot 24.413927909925626^2 \cdot 0.03 \cdot 1.2 \\
&= 12.916950012575231 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 12.916950012575231 \text{ Newtons in direction away from } 0.038126740261677436 \\
&= (-12.907562798438319, -0.4923618910297632)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -600.0 \cdot 9.8) \\
&= (0, -5880.0)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (375.84364312296117, 77.27853054309396) + (-643.7399680160003, 5734.823730964559) + \\
&\quad (1.7832633012530916, -68.23769642914075) + (-12.907562798438319, -0.4923618910297632) + \\
&\quad (0, -5880.0) + (0, 0) \\
&= (-279.0206243902243, -136.62779681251686)
\end{aligned}$$

Linear Acceleration

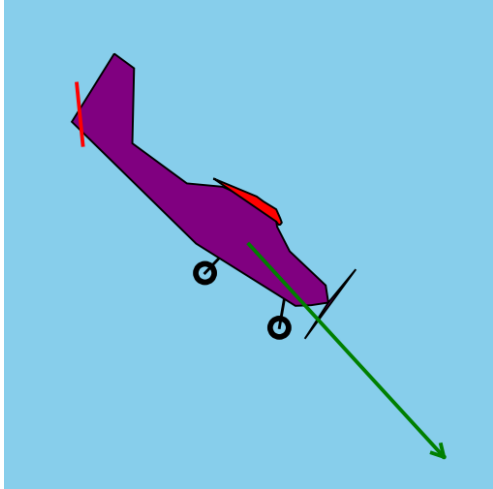
$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(-279.0206243902243, -136.62779681251686)}{600.0} \\&= (-0.4650343739837072, -0.22771299468752812)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (24.396185426677608, 0.9305979895790267) + (-0.4650343739837072, -0.22771299468752812) \cdot 0.01 \\&= (24.39153508293777, 0.9283208596321515) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (1912.6724958372301, 180.34367644747098) + (24.39153508293777, 0.9283208596321515) \cdot 0.01 \\&= (1912.9164111880596, 180.3529596560673)\end{aligned}$$

11 Test 10

```
test_aircraft10 = Cessna172(angular_velocity=2.5462461874185127, engine_rpm=5789.881024080416,  
    linear_velocity=(39.59202073532071, -43.20742419924472), pitch_angle_radians=-0.6335645532198284,  
    position=(1886.9970724837021, 388.531086940641), stabilizer_angle_radians=-0.2792526803190927, throttle=1.0,  
    time_elapsed=67.196)  
simulate_one_time_step(test_aircraft10, 0.01)
```



Engine torque generated

$$\begin{aligned}\tau_e &= \text{Cessna172.engine_torque}(\text{throttle}, N_e) \\ &= \text{Cessna172.engine_torque}(1.0, 5789.881024080416) \\ &= -374.9671959752841\end{aligned}$$

Airspeed

$$\begin{aligned}|v| &= |(39.59202073532071, -43.20742419924472)| \\ &= 58.60383615293066\end{aligned}$$

Engine RPM

$$N_e = 5789.881024080416$$

Propeller RPM

$$\begin{aligned} N_p &= \frac{N_e}{G} \\ &= \frac{5789.881024080416}{1.0} \\ &= 5789.881024080416 \end{aligned}$$

Advance ratio

$$\begin{aligned} J &= \frac{|v|}{\frac{N_p}{60} \cdot D_p} \\ &= \frac{58.60383615293066}{\frac{5789.881024080416}{60} \cdot 1.88} \\ &= 0.32303516921872943 \end{aligned}$$

Air Density

$$\begin{aligned} h &= 388.531086940641 \\ T_h &= T_0 - L \cdot h \\ &= 288.15 - 0.0065 \cdot 388.531086940641 \\ &= 285.6245479348858 \\ P_h &= P_0 \cdot \left(\frac{T_h}{T_0} \right)^{\frac{g}{L \cdot R}} \\ &= 101325 \cdot \left(\frac{285.6245479348858}{288.15} \right)^{\frac{9.8}{0.0065 \cdot 287.05}} \\ &= 96746.7345806474 \\ \rho &= \frac{P_h}{R \cdot T_h} \\ &= \frac{96746.7345806474}{287.05 \cdot 285.6245479348858} \\ &= 1.1800033337431777 \end{aligned}$$

Propeller radians per second

$$\begin{aligned} \omega_p &= \frac{N_p \cdot 2 \cdot \pi}{60} \\ &= \frac{5789.881024080416 \cdot 2 \cdot 3.141592653589793}{60} \\ &= 606.3149230136661 \end{aligned}$$

Propeller torque load

$$\begin{aligned}
Q_p &= C_Q(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^5 \\
&= \text{Cessna172.propeller_torque_coefficient}(0.32303516921872943) \cdot 1.1800033337431777 \cdot +606.3149230136661^2 \cdot 1.88^5 \\
&= 0.0001753041725751522 \cdot 1.1800033337431777 \cdot +606.3149230136661^2 \cdot 1.88^5 \\
&= 1785.9169076840767
\end{aligned}$$

Engine torque load

$$\begin{aligned}
Q_e &= \frac{Q_p}{G} \\
&= \frac{1785.9169076840767}{1.0} \\
&= 1785.9169076840767
\end{aligned}$$

Engine net torque

$$\begin{aligned}
\tau_{\text{net}} &= \tau_e - Q_e \\
&= -374.9671959752841 - 1785.9169076840767 \\
&= -2160.884103659361
\end{aligned}$$

Engine angular acceleration

$$\begin{aligned}
\alpha_e &= \frac{\tau_{\text{net}}}{I_e} \\
&= \frac{-2160.884103659361}{0.2} \\
&= -10804.420518296803
\end{aligned}$$

Update Engine RPM

$$\begin{aligned}
N_e &= N_e + \alpha_e \cdot \Delta t \cdot \frac{60}{\pi} \\
&= 5789.881024080416 + -10804.420518296803 \cdot 0.01 \cdot \frac{60}{3.141592653589793} \\
&= 3726.3887048038946 \\
&= 2700 \text{ reduced to max Cessna172 engine RPM}
\end{aligned}$$

Wing moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to wing} \\
&= -0.6335645532198284 + 1.6406094968746698 \\
&= 1.0070449436548414
\end{aligned}$$

Angle of Wings

$$\begin{aligned}
\text{angle of wings} &= \theta + \text{angle of wings (relative to aircraft)} \\
&= -0.6335645532198284 + 0.026179938779914945 \\
&= -0.6073846144399134
\end{aligned}$$

Angle to Relative Airflow

$$\begin{aligned}
RAF &= \text{Angle to } v \text{ (relative to horizontal)} \\
&= \text{Angle to } (39.59202073532071, -43.20742419924472) \\
&= -0.8290350306135054
\end{aligned}$$

Angle of Attack (wing)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (wing)} - \text{angle to relative airflow} \\
&= -0.6073846144399134 - -0.8290350306135054 \\
&= 0.22165041617359194
\end{aligned}$$

Magnitude of Lift (wing)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 16.2 \cdot \text{Cessna172.wing_lift_coefficient}(0.22165041617359194) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 16.2 \cdot 1.569963337406506 \\
&= 51535.8988094781
\end{aligned}$$

Lift Force (wing)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= 51535.8988094781 \text{ in direction normal to } -0.8290350306135054 \\
&= 51535.8988094781 \text{ in direction } 0.7417612961813912 \\
&= (37996.37681635141, 34817.01042494314)
\end{aligned}$$

Induced Drag Magnitude(wing)

$$\begin{aligned} |F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\ &= \frac{51535.8988094781^2}{0.5 \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 16.2 \cdot 3.141592653589793 \cdot +0.8 \cdot AR} \\ &= 4310.117898226902 \end{aligned}$$

Induced Drag (wing)

$$\begin{aligned} F_i &= |F_i| \text{ in direction away from relative airflow} \\ &= 4310.117898226902 \text{ in direction away from } -0.8290350306135054 \\ &= (-2911.861891650974, 3177.762832649205) \end{aligned}$$

Total Aerodynamic Force (wing)

$$\begin{aligned} F_a &= F_L + F_i \\ &= (37996.37681635141, 34817.01042494314) + (-2911.861891650974, 3177.762832649205) \\ &= (35084.51492470044, 37994.773257592344) \end{aligned}$$

Force component normal to moment arm (wing)

$$\begin{aligned} |F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\ &= \text{component of } (35084.51492470044, 37994.773257592344) \text{ in direction } 1.0070449436548414 + 90^\circ \\ &= \text{component of } (35084.51492470044, 37994.773257592344) \text{ in direction } 2.577841270449738 \\ &= -9352.49465022511 \end{aligned}$$

Compute Moment(wing)

$$\begin{aligned} M &= |F_n| \cdot d \\ &= -9352.49465022511 \cdot 1.0 \\ &= -9352.49465022511 \end{aligned}$$

Stabilizer moment arm

$$\begin{aligned}
\gamma &= \theta + \text{angle from center of gravity to stabilizer} \\
&= -0.6335645532198284 + 3.0543261909900767 \\
&= 2.4207616377702483
\end{aligned}$$

Angle of Stabilizer

$$\begin{aligned}
\text{angle of stabilizer} &= \theta + \text{angle of stabilizer (relative to aircraft)} \\
&= -0.6335645532198284 + -0.2792526803190927 \\
&= -0.9128172335389211
\end{aligned}$$

Angle of Attack (stabilizer)

$$\begin{aligned}
\phi &= \text{angle of aerofoil (stabilizer)} - \text{angle to relative airflow} \\
&= -0.9128172335389211 - -0.8290350306135054 \\
&= -0.08378220292541572
\end{aligned}$$

Magnitude of Lift (stabilizer)

$$\begin{aligned}
|F_L| &= \frac{1}{2} \cdot \rho \cdot |v|^2 \cdot S \cdot C_L(\phi) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 3.0 \cdot \text{Cessna172.stabilizer_lift_coefficient}(-0.08378220292541572) \\
&= \frac{1}{2} \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 3.0 \cdot -0.38402933007479517 \\
&= -2334.4844150947397
\end{aligned}$$

Lift Force (stabilizer)

$$\begin{aligned}
F_L &= |F_L| \text{ in direction normal to relative airflow} \\
&= -2334.4844150947397 \text{ in direction normal to } -0.8290350306135054 \\
&= -2334.4844150947397 \text{ in direction } 0.7417612961813912 \\
&= (-1721.1681867771372, -1577.1485526565118)
\end{aligned}$$

Induced Drag Magnitude(stabilizer)

$$\begin{aligned}
|F_i| &= \frac{|F_L|^2}{0.5 \cdot \rho \cdot |v|^2 \cdot A_w \cdot \pi \cdot e \cdot AR} \\
&= \frac{-2334.4844150947397^2}{0.5 \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 3.0 \cdot 3.141592653589793 \cdot +0.7 \cdot AR} \\
&= 94.36777472251299
\end{aligned}$$

Induced Drag (stabilizer)

$$\begin{aligned}
F_i &= |F_i| \text{ in direction away from relative airflow} \\
&= 94.36777472251299 \text{ in direction away from } -0.8290350306135054 \\
&= (-63.75369154691364, 69.57545343165184)
\end{aligned}$$

Total Aerodynamic Force (stabilizer)

$$\begin{aligned}
F_a &= F_L + F_i \\
&= (-1721.1681867771372, -1577.1485526565118) + (-63.75369154691364, 69.57545343165184) \\
&= (-1784.9218783240508, -1507.57309922486)
\end{aligned}$$

Force component normal to moment arm (stabilizer)

$$\begin{aligned}
|F_n| &= \text{component of } F_a \text{ in direction normal to moment arm angle } \gamma \\
&= \text{component of } (-1784.9218783240508, -1507.57309922486) \text{ in direction } 2.4207616377702483 + 90^\circ \\
&= \text{component of } (-1784.9218783240508, -1507.57309922486) \text{ in direction } 3.991557964565145 \\
&= 2310.6404863303915
\end{aligned}$$

Compute Moment(stabilizer)

$$\begin{aligned}
M &= |F_n| \cdot d \\
&= 2310.6404863303915 \cdot 4.84 \\
&= 11183.499953839095
\end{aligned}$$

Net Moment

$$\begin{aligned}M_{net} &= M_w + M_s \\&= -9352.49465022511 + 11183.499953839095 \\&= 1831.0053036139852\end{aligned}$$

Angular Acceleration

$$\begin{aligned}\alpha_t &= \frac{M_{net}}{I_p} \\&= \frac{1831.0053036139852}{1200.0} \\&= 1.5258377530116543\end{aligned}$$

Update Angular Velocity and Pitch

$$\begin{aligned}\omega_{(t+\Delta t)} &= \omega_t + \alpha_{(t+\Delta t)} \cdot \Delta t \\&= 2.5462461874185127 + 1.5258377530116543 \cdot 0.01 \\&= 2.5615045649486294 \\ \theta_{(t+\Delta t)} &= \theta_t + \omega_{(t+\Delta t)} \cdot \Delta t \\&= -0.6335645532198284 + 2.5615045649486294 \cdot 0.01 \\&= -0.6079495075703422\end{aligned}$$

Engine Thrust Force

$$\begin{aligned}
|F_t| &= C_T(J) \cdot \rho \cdot \omega_p^2 \cdot D_p^4 \\
&= \text{Cessna172.propeller_thrust_coefficient}(0.6927167393963436) \cdot 1.1800033337431777 \cdot 282.7433388230814^2 \cdot 1.88^4 \\
&= 0.0015604305568802947 \cdot 1.1800033337431777 \cdot 282.7433388230814^2 \cdot 1.88^4 \\
&= 1838.839574478919 \\
F_t &= |F_t| \text{ in direction } \theta \\
&= 1838.839574478919 \text{ Newtons, in direction } -0.6079495075703422 \\
&= (1509.3580543286753, -1050.3186395103953)
\end{aligned}$$

Parasitic Drag Force

$$\begin{aligned}
|F_d| &= 0.5 \cdot \rho \cdot |v|^2 \cdot C_d \cdot A_{\text{plane}} \\
&= 0.5 \cdot 1.1800033337431777 \cdot 58.60383615293066^2 \cdot 0.025 \cdot 1.5 \\
&= 75.98652733894264 \\
F_d &= |F_d| \text{ in direction away from angle to relative airflow} \\
&= 75.98652733894264 \text{ Newtons in direction away from } -0.8290350306135054 \\
&= (-51.33555008511134, 56.02333116203375)
\end{aligned}$$

Weight Force

$$\begin{aligned}
F_g &= (0, -m \cdot g) \\
&= (0, -1086.0 \cdot 9.8) \\
&= (0, -10642.800000000001)
\end{aligned}$$

Rolling Resistance Force

$$F_r = (0, 0) \text{ since not on ground or not moving}$$

Net Force

$$\begin{aligned}
F_{\text{net}} &= F_t + F_w + F_s + F_d + F_g + F_r \\
&= (1509.3580543286753, -1050.3186395103953) + (38066.634225493974, 41871.436721098806) + \\
&\quad (-1225.6786516277175, -1061.4252767114413) + (-51.33555008511134, 56.02333116203375) + \\
&\quad (0, -10642.800000000001) + (0, 0) \\
&= (38298.978078109816, 29172.91613603899)
\end{aligned}$$

Linear Acceleration

$$\begin{aligned}a_t &= \frac{F_{\text{net}}}{m} \\&= \frac{(38298.978078109816, 29172.91613603899)}{1086.0} \\&= (35.26609399457626, 26.86272204055156)\end{aligned}$$

Update Linear Velocity and Position

$$\begin{aligned}v_{(t+\Delta t)} &= v_t + a_{(t+\Delta t)} \cdot \Delta t \\&= (39.59202073532071, -43.20742419924472) + (35.26609399457626, 26.86272204055156) \cdot 0.01 \\&= (39.94468167526647, -42.93879697883921) \\r_{(t+\Delta t)} &= r_t + v_{(t+\Delta t)} \cdot \Delta t \\&= (1886.9970724837021, 388.531086940641) + (39.94468167526647, -42.93879697883921) \cdot 0.01 \\&= (1887.3965193004549, 388.1016989708526)\end{aligned}$$